Low-Volatility Cycles Revisited

Replication, Post-Sample Evidence and the Implementation of Low-Beta Investing

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Abstract

Can investors build a robust, implementable strategy on the low-beta anomaly? Garcia-Feijóo et al. (2015, hereafter GFKSW) argue that it is cyclical: low-beta stocks beat high-beta stocks only when they start out cheap, and value and momentum absorb much of the conditional alpha. Rebuilding their study from CRSP and Compustat, we reproduce their unconditional results almost exactly and the zero-cost portfolio's one-year returns after the extreme valuation states, but not the middle states. The cheap-minus-expensive contrasts are not robustly significant at the 5% level. A bias-corrected predictive regression, which tests a different hypothesis (a linear relation between spread and next-year return), indicates in sample that the spread predicts the beta-neutral portfolio's return ($t = 2.95$). The conditional premium coincides with a value exposure; its four-factor alphas are insignificant under our bootstraps. Since 2013, valuations have never been cheap enough to test the central prediction. For long-only investors, the 65% of the zero-cost portfolio's CAPM alpha that comes from shorting high-beta stocks is out of reach: low-beta portfolios cut volatility by about a fifth but do not beat the cap-weighted benchmark significantly (net Sharpe ratio 0.50 against 0.45, $p = 0.53$), and the daily nonlinear-shrinkage minimum-variance portfolio has the lowest realized volatility of all portfolios but, after costs, a net Sharpe ratio no higher than the simple low-beta quintile's.

1 Introduction

Since Black et al. (1972) documented that the security market line is too flat and Black (1972) explained it with borrowing restrictions, a large literature has shown that low-risk stocks earn risk-adjusted returns above, and raw returns at least equal to, those of high-risk stocks (Ang et al. 2006; Blitz and van Vliet 2007; Baker et al. 2011; Frazzini and Pedersen 2014). The finding contradicts the most basic prediction of the CAPM and underlies minimum-volatility indices and low-beta funds; whether investors can capture it in a robust, implementable strategy is our question.

Garcia-Feijóo et al. (2015) qualify this evidence on three counts. First, the conventional zero-cost portfolio that is long the lowest-beta and short the highest-beta quintile carries a market beta of about −1.3; shorting only a quarter of the high-beta quintile removes most of this exposure. Second, the premium is cyclical: the lowest-risk portfolio outperforms "only when valuation is strongly in its favor" (Garcia-Feijóo et al. 2015, 54), that is, when the book-to-price ratio of low-beta stocks is high relative to that of high-beta stocks by historical standards. Third, much of the conditional CAPM alpha disappears once value and momentum are controlled for. They do not propose a timing rule, and their sample ends in December 2012.

We ask whether the valuation dependence of the low-beta premium is a robust feature that an investor can use or an in-sample pattern resting on a few episodes. We answer in three steps: we rebuild GFKSW's study from the raw data and test its central contrast with inference that respects overlapping windows; we follow the strategies to 2024; and we ask what an investor can capture once beta has to be estimated, short sales are restricted, trading costs are paid and performance is judged at daily frequency and over rolling windows.

The paper makes three contributions. First, it separates what replicates from what does not: GFKSW's unconditional results and the one-year returns after the extreme states reproduce, their middle states do not, and, in our reconstruction and simulations, their Newey-West statistics overstate the precision of averages over overlapping windows. The cheap-minus-expensive contrasts are not robustly significant at the 5% level. That of the zero-cost portfolio rests largely on the technology bust and the 2009 rebound, and the conditional premium coincides largely with a value exposure. A bias-corrected predictive regression, testing the different hypothesis of a linear relation between spread and next-year return, finds in sample that the spread predicts the beta-neutral portfolio's return. Second, a CAPM decomposition attributes the negative post-sample excess returns of the zero-cost portfolio mainly to its negative market beta in a rising market, while valuations never reached the states in which GFKSW expect low-beta stocks to win. Third, it quantifies the gap between the academic factor and an investable portfolio: most of the premium comes from the short side, a long-only low-beta portfolio offers lower risk but no significant net Sharpe ratio gain, better beta forecasts bring no significant gain in gross Sharpe ratios, and the daily nonlinear-shrinkage portfolio has the lowest realized volatility but no higher net Sharpe ratio.

Sections 2 and 3 cover literature and methods, Section 4 the replication, Section 5 post-sample performance and timing, Sections 6 to 8 the implementation, and Section 9 concludes.

2 Related literature

Risk-based explanations of the low-risk anomaly build on Black (1972): investors who cannot or will not borrow tilt towards high-beta stocks, which flattens the security market line (Frazzini and Pedersen 2014). Institutional explanations stress benchmarking: managers measured against a

cap-weighted index have little incentive to hold low-beta stocks that raise tracking error (Baker et al. 2011). Low-risk investing works within and across industries (Asness et al. 2014), limits to arbitrage concentrate the anomaly in small, illiquid stocks (Li et al. 2014), one reason GFKSW restrict attention to large caps, and much of the betting-against-beta premium depends on construction choices such as beta shrinkage and equal weighting (Novy-Marx and Velikov 2022). The role of value and momentum relates to the factor models of Fama and French (1993) and Carhart (1997) and to the fragility of momentum (Daniel and Moskowitz 2016); timing factors with their own valuation spreads has been hard in practice (Asness et al. 2017).

For the implementation analyses, Hollstein et al. (2019) find that betas from daily returns forecast future betas better than five-year monthly estimates, and Dimson (1979) corrects daily betas for non-synchronous trading. Many anomalies do not survive trading costs (Novy-Marx and Velikov 2016; De Nard and Zhao 2022), long-only implementations can be preferable once costs and constraints are considered (Blitz et al. 2014), the $1/N$ portfolio is a hard benchmark (DeMiguel et al. 2009), and shrinkage makes minimum-variance portfolios usable in large cross-sections (Ledoit and Wolf 2004, 2020; Clarke et al. 2006). Published anomalies weaken after publication (McLean and Pontiff 2016), significance thresholds should rise with the number of tests (Harvey et al. 2016), and carefully replicated factors largely survive (Jensen et al. 2023).

3 Data and methodology

3.1 Data

Monthly and daily returns of US common stocks traded on NYSE, AMEX and NASDAQ come from CRSP via the university's Tidy Finance database (Scheuch et al. 2023), in CRSP's current (CIZ) format. Holding-period returns come from CRSP's monthly security file, which keeps the month in which a stock delists and includes the delisting return in that month's total return; stocks whose last trading day falls in the formation month are not bought. Every held stock therefore has a return.[1] GFKSW instead set the delisting return, not the whole month's return, of performance-related delistings to −30% (NYSE, AMEX) or −55% (NASDAQ), following Shumway (1997). In our large-cap universe, a held stock was dropped by its exchange in only 6 stock-months of 1968–2012 (25 over 1929–2024), one of them in an extreme beta quintile, so applying their rule leaves the mean low-minus-high return unchanged to within 0.01 basis points (Appendix Table 10). In our sample, the delisting treatments tested do not explain differences from their results.

Book equity follows Fama and French (1992) and uses only December fiscal year-ends with at least two years of Compustat history, as in GFKSW; before Compustat coverage we use the book equity of Davis et al. (2000), GFKSW's source for the years before 1951. Factors and the one-month T-bill rate are from Kenneth French's data library (French 2026); GFKSW use three-month T-bills. Instead of the CRSP value-weighted index, we use the Fama–French market total return (MKT–RF + RF), whose mean over 1968–2012 matches our panel's value-weighted return (both 0.870% per month; correlation 1.000). Regressions use the market excess return, MKT–RF. The momentum factor ends in December 2024, which ends every sample. Daily betas use CRSP's daily returns; the daily portfolio paths of Section 8 also use the delisting-day records of CRSP's daily file. GFKSW used the CRSP vintage of 2013, whose effect cannot be quantified here. Appendix Table 8 compares the two designs.

[1] Over 1968–2012, delisting months carry on average 0.13% to 0.19% of a beta quintile's weight, the formation-month exclusions 0.03% to 0.05%. Dropping the delisting months instead changes the mean low-minus-high return by 0.04 basis points per month (Appendix Table 10).

3.2 Portfolios and return definitions

At the end of each month t, each stock's CAPM beta is estimated from its excess returns over the previous 60 months (at least 24, including month t). The universe consists of stocks priced between USD 2 and USD 1,000 with a market capitalisation above the 67th percentile of NYSE stocks, on average 565 stocks per month in 1968–2012. Stocks are sorted into beta quintiles and held, value-weighted, over month $t+1$; all portfolios are rebalanced monthly unless stated otherwise.

All returns are total returns in USD; the excess return of a portfolio is its return minus the one-month T-bill rate r_f, and its active return is its return minus that of a benchmark. The low-minus-high portfolio (L–H) holds the investor's capital as collateral in T-bills and is long USD 1 of quintile 1 for every USD 1 of quintile 5 sold short, the short proceeds financing the long position. Its total return is $r_f + r_{Q1} - r_{Q5}$ and its excess return $r_{Q1} - r_{Q5}$. GFKSW's beta-neutral portfolio (BN) invests the capital in quintile 1 and shorts quintile 5 in an amount of 25% of capital, with the short proceeds in T-bills. Its total return is $r_{Q1} - 0.25\, r_{Q5} + 0.25\, r_f$ and its excess return $(r_{Q1} - r_f) - 0.25\,(r_{Q5} - r_f) = r_{Q1} - 0.25\, r_{Q5} - 0.75\, r_f$; the $0.75\, r_f$ term is the opportunity cost of the fully invested long position net of the interest on the short proceeds. Like GFKSW (their note 6), Table 1 reports the raw return $r_{Q1} - 0.25\, r_{Q5}$ and computes Sharpe ratios from excess returns. These gross returns ignore lending fees and the spread between the rebate on short proceeds and the T-bill rate; net returns are defined in Section 3.4. The fixed short ratio of 25% matches GFKSW's full-sample average betas and therefore uses future information; we also reset it every month to $\hat{\beta}_{Q1,t}/\hat{\beta}_{Q5,t}$ (BN ex ante) and, as Frazzini and Pedersen (2014) do, to the same ratio with betas shrunk towards one, $0.6\hat{\beta} + 0.4$ (BN shrunk).

For horizons of h months we compound the monthly excess returns of the monthly rebalanced strategy, $\prod_{k=1}^{h}(1 + x_{t+k}) - 1$, and annualise geometrically. This compounds monthly excess returns and is distinct from the total-return wealth paths in Figure 1; GFKSW rebalance monthly (their notes 9 and 10) but do not state how they compound. Relative wealth of two buy-and-hold portfolios and the payoff of a never-rebalanced long-short position answer different questions (Appendix Table 12).

3.3 Valuation states and inference

The B/P spread is the average book-to-price ratio of quintile 1 minus that of quintile 5, with December book equity of year $y-1$ used from the end of June of year y and market value measured at the end of the formation month (Asness and Frazzini 2013). The higher the spread, the cheaper low-beta stocks are relative to high-beta stocks; because the spread is positive in most months, "cheap" and "expensive" below always mean relative to the spread's own history. GFKSW "deleted the top and bottom 0.5% of the B/P observations" but report 540 months with 108 per state; we therefore read the trimming as applying to stock-level ratios, not months, and trim each month's cross-section of large-cap B/P ratios at the 0.5th and 99.5th percentiles, which uses no future information. The 540 formation months from December 1967 to November 2012 (return months January 1968 to December 2012) are sorted into spread quintiles ("states"): state 5 is the cheapest fifth for low-beta stocks, state 1 the most expensive. As in GFKSW, these breakpoints use the whole sample; Section 5 also uses breakpoints estimated in real time. Only windows that end by December 2012 enter the replication (state 1 keeps 97 of 108 windows at one year and 64 at five years).

Tables 2 and 3 average, for every state, a statistic computed over the h months after each formation month: the annualised chained excess return, or the intercept of a CAPM or four-factor regression. Where we replicate GFKSW's tables we also report their statistic as we read it, a Newey-West t

with h lags on the state-filtered sequence of window estimates. Overlapping windows and states that cluster in a few episodes make that sequence strongly dependent. Our baseline is therefore a block bootstrap. We cut the monthly history of strategy and factor returns into blocks of 24 months, draw blocks with replacement, wrapping around the end of the sample (Künsch 1989; Politis and Romano 1992), keep the dates of the valuation signals fixed and recompute every state average (999 draws). Because the reshuffled returns are unrelated to the signals, the draws show how far apart the state averages fall by chance when the states carry no information; their standard deviation gives the t-statistics and their centred quantiles the p-values, not confidence intervals. Blocks of 24 months lie between the 3 months that the rule of (Politis and White 2004; corrected by Patton et al. 2009) selects for the mean of the L–H returns and the 38 months it selects for their squares; Appendix Table 18 reports blocks of 12 to 60 months.

The bootstrap assumes that volatility is unrelated to the signal dates, which fails here: in the year after a state-5 month the L–H volatility averages 28% per year, against 17% to 19% after the other states. A volatility-preserving variant therefore rescales the reshuffled, standardised returns with the local volatility of the original dates. Size simulations with the actual state calendar (Appendix Table 19) show that the 24-month bootstrap rejects a true null at the 5% level in 3.6% to 6.0% of the samples when volatility is unrelated to the calendar, the GFKSW-style statistic in 12.6% to 14.6%. Under the observed volatility path the baseline bootstrap over-rejects, most for the state-5 means (Appendix Table 19); for the L–H contrast its size-adjusted 5% critical value is 2.29, that of the volatility-preserving variant 2.10. No procedure can overcome the small number of episodes. Calendar-time regressions on stacked windows estimate a different quantity and are reported in Appendix Table 15.

A predictive regression of the next year's return on the standardised spread avoids classifying months into states. The spread is persistent and its innovations correlate with returns, inducing small-sample OLS bias (Stambaugh 1999); we therefore report the reduced-bias estimator of Amihud and Hurvich (2004) on non-overlapping December observations.

The extensions use Newey-West t-statistics with the rule-of-thumb lag $\lfloor 4(T/100)^{2/9} \rfloor$ (Newey and West 1987). Sharpe ratios are compared with the HAC test of Ledoit and Wolf (2008) on the common sample, beta forecasts with Diebold-Mariano tests, and within each family of comparisons with a common benchmark we report Holm-adjusted p-values (Harvey et al. 2016). Three analyses are primary: the replication, the performance after 2012, and the comparison of the long-only low-beta quintile with the cap-weighted benchmark after costs (one test, unadjusted). Everything else is exploratory.

3.4 Costs, turnover and liquidity

Turnover at a rebalancing date is $\sum_i |w_{i,t} - \tilde{w}_{i,t}|$, where $\tilde{w}$ are the previous target weights after drifting with returns over the holding period. It counts purchases and sales, so a one-way cost c per dollar traded is charged as c times turnover; a stock that delisted during the month counts as a sale. Every portfolio, including each benchmark, bears costs on its own turnover. The base case of $c = 10$ basis points is the flat cost used by De Nard and Zhao (2022) and twice the cost they assume for large caps; because we have no transaction-level data, 0 to 50 basis points are reported as scenarios. Short positions pay a lending fee of 25 basis points per year on the short notional, the value-weighted borrowing cost of the loan portfolio studied by D'Avolio (2002) (17 basis points for the 91% of lent stocks that are cheap to borrow); this, too, is a scenario value, not calibrated to our sample. The cost model ignores market impact, financing spreads and the possibility that a stock cannot be borrowed. CRSP reports trading volume, which we use only to describe the

liquidity of each portfolio (Section 7); we have no data on market impact or borrowing availability.

4 Replication

4.1 Risk, return and the construction of the long-short portfolio

Table 1: Risk and return of beta-sorted portfolios, January 1968 to December 2012 (replication of GFKSW Table 1)

	Arithmetic mean		Geometric mean		Beta			Sharpe ratio	
Beta quintile	Ours	GFKSW	Ours	GFKSW	Ours	GFKSW	Realised	Ours	GFKSW
1 (low)	0.89	0.89	0.82	0.83	0.47	0.47	0.63	0.43	0.45
2	0.94	0.92	0.85	0.84	0.77	0.76	0.81	0.42	0.42
3	0.87	0.86	0.75	0.75	0.99	0.98	0.97	0.32	0.32
4	0.85	0.85	0.70	0.70	1.24	1.24	1.14	0.27	0.27
5 (high)	0.73	0.73	0.46	0.47	1.76	1.76	1.45	0.14	0.15
Low – high	0.16 (0.62)	0.16	−0.02	−0.02	−1.29	−1.29	−0.82	0.09	0.10
Beta neutral	0.70 (5.57)	0.71	0.66	0.67	0.03	0.03	0.27	0.45	0.46
Market	0.87 (4.36)	0.87	0.76	0.76	1.00	1.00	1.00	0.33	0.33

Value-weighted monthly raw returns in percent of large-cap stocks sorted on 60-month CAPM betas; ordinary t-statistics in parentheses, as in GFKSW. Beta: average pre-formation beta; Realised: slope of the portfolio's monthly excess return on the market excess return. Sharpe ratios are annualised and use excess returns (L–H: $r_{Q1} - r_{Q5}$; beta neutral: $r_{Q1} - 0.25\, r_{Q5} - 0.75\, r_f$). The beta-neutral mean is the raw return $r_{Q1} - 0.25\, r_{Q5}$. 540 months; GFKSW values from their Table 1.

Against the published values (Table 1), the average absolute difference in arithmetic means across the eight portfolios is 0.6 basis points per month, the pre-formation betas differ by at most 0.01 and the Sharpe ratios by at most 0.02, and the t-statistics of the two long-short portfolios (0.62 and 5.57) are close to GFKSW's 0.63 and 5.49. Average returns are flat across the first four quintiles and fall in the highest-beta quintile, while Sharpe ratios fall from 0.43 to 0.14. The L–H portfolio's mean excess return is 0.16% per month and statistically insignificant; its market beta is −1.29 ex ante and −0.82 ex post. The beta-neutral portfolio earns 0.38% per month in excess of T-bills ($t = 2.87$, Table 4), the number comparable with L–H; the raw return of 0.70% and its t of 5.57, which Table 1 reports to match GFKSW, also contain three quarters of the T-bill return.

Neither portfolio is what its name suggests. Ex post, the quintile betas are compressed towards one, from 0.47 to 0.63 for quintile 1 and from 1.76 to 1.45 for quintile 5, as estimation error and mean reversion in beta imply. The beta-neutral portfolio therefore has a realised beta of 0.27 (Newey-West standard error 0.04), and part of its return is ordinary equity premium. Shrinking the betas towards one before setting the short ratio, as Frazzini and Pedersen (2014) do, raises the short position to 48% of capital on average and lowers the realised beta to −0.06 (1968–2024: −0.06, Appendix Table 23). Restricting the universe to firms with a December book value barely changes the extreme quintiles and the long-short portfolios of Table 1, although it reorders the mean returns of the middle quintiles (Appendix Table 14).

Figure 1 shows how the construction shapes realised wealth. One dollar invested in January 1968 grew to 281 dollars in the low-beta quintile, 78 in the high-beta quintile and 306 in the market by December 2024. The dollar-neutral L–H portfolio ended at 3.7 dollars, below the 12.2 of T-bills; the beta-neutral portfolio ended at 132 dollars. Appendix Table 23 reports their mean excess returns and CAPM decompositions.

4.2 Valuation cycles

Figure 2 shows the valuation cycle that defines GFKSW's states (their Figures 3 and 5). The spread is positive in 86% of the 540 formation months between December 1967 and November

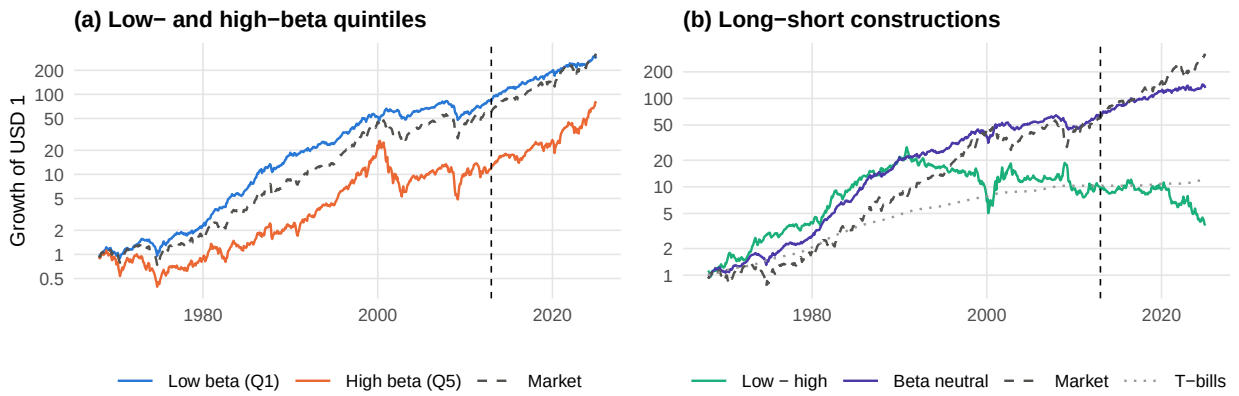


Figure 1: Cumulative growth of USD 1 (log scale), January 1968 to December 2024, from monthly gross total returns, plotted at the end of each return month. Panel (a): value-weighted lowest- and highest-beta quintiles of the large-cap universe and the Fama-French market. Panel (b): the low-minus-high portfolio (capital in T-bills, long Q1 and short Q5: $r_f + r_{Q1} - r_{Q5}$), the beta-neutral portfolio ($r_{Q1} - 0.25\, r_{Q5} + 0.25\, r_f$), the market and T-bills. Dashed vertical line: end of the GFKSW sample (December 2012).

2012, so high-beta stocks normally trade at a valuation premium. The spread was negative in the early 1970s and, from 2008 onwards, in most months of the following decade. Its largest reversal occurred during the Great Depression: in the median month of the 1930s, the average B/P ratio of high-beta stocks was 2.7 times that of low-beta stocks (Appendix Figure 5). At the end of December 2012, when GFKSW's sample ends, the spread stood at −0.09, at the 8th percentile of the 540 formation months of their sample, and it has stayed below the upper two breakpoints since. The cheapest states for low-beta stocks occurred mainly in 1978–1981, 1988 and 1999–2000, the most expensive ones in 1970–1974 and 2005–2012.

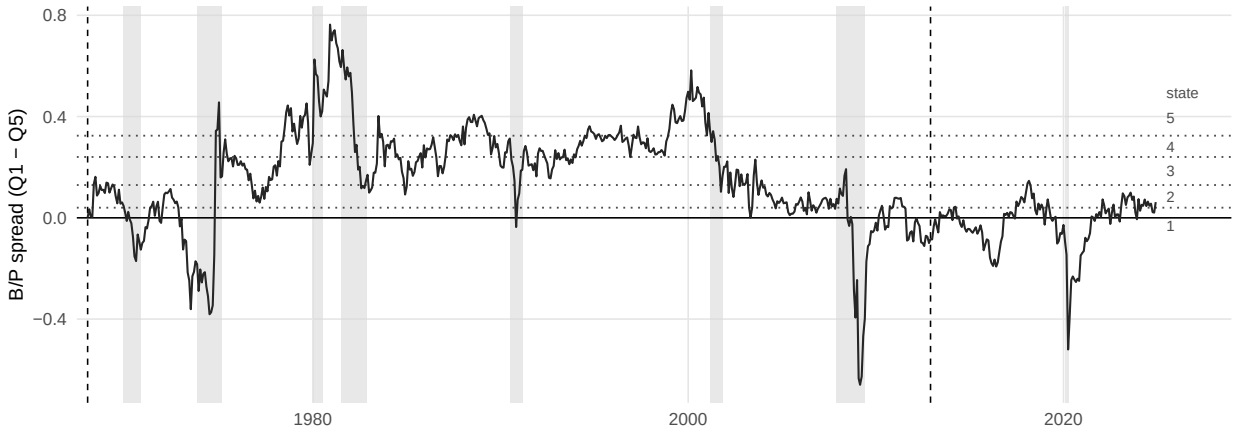


Figure 2: B/P spread of the beta quintiles (average book-to-price ratio of the low-beta minus that of the high-beta quintile; stock-level B/P trimmed cross-sectionally at 0.5% in each tail), measured at the end of each formation month from December 1967 to November 2024. Dotted horizontal lines: breakpoints between the five valuation states, estimated over the GFKSW formation months (state 1 below the lowest line, state 5 above the highest). Dashed vertical lines: first and last formation month of the GFKSW sample (December 1967, November 2012). Shaded: NBER recessions. The B/P levels since 1929 are shown in Figure 5.

Table 2 reproduces GFKSW's pattern at the extremes. When low-beta stocks start cheapest (state 5), L–H earns 11.8% over the following year (GFKSW: 12.52%); when they start most expensive (state 1), −4.9% (GFKSW: −5.80%). The beta-neutral portfolio earns 10.3% after state 5 (GFKSW: 10.27%), but only 0.9% after state 1 (GFKSW: 2.30%), and unlike GFKSW's, its average return is negative at two and three years after state 2. The middle states do not replicate. Our state cut-offs lie above GFKSW's (0.04, 0.13, 0.24, 0.32 against −0.01, 0.12, 0.22, 0.27),

Table 2: Annualised future cumulative excess returns by initial B/P spread state, 1968–2012 (GFKSW Table 2, Panels B and C)

	This replication (%)					Bootstrap t		GFKSW (%)	
	Horizon (months)					12 months		Horizon	
State	12	24	36	48	60	Baseline	Vol.-pres.	12	60
Panel A: low-minus-high (L–H)									
1 (low)	−4.93	−1.19	1.04	1.29	1.56	−0.80	−0.91	[−5.80]	[−2.83]
2	7.33	1.83	−0.25	0.09	1.28	1.35	1.73	[4.64]	[0.24]
3	−1.60	−2.35	−2.89	−1.66	−1.13	−0.31	−0.42	[−6.84]	[−2.23]
4	−2.99	−6.17	−4.62	−4.01	−3.72	−0.58	−0.66	[3.36]	[−2.81]
5 (high)	11.79	10.67	6.48	3.93	2.45	2.01	1.47	[12.52]	[4.40]
Panel B: beta neutral (BN)									
1 (low)	0.88	2.61	2.62	0.47	0.92	0.26	0.25	[2.30]	[1.62]
2	1.73	−0.52	−0.18	1.32	1.42	0.57	0.72	[3.97]	[2.61]
3	4.19	4.13	4.66	5.23	5.68	1.48	1.63	[2.53]	[3.49]
4	6.19	4.99	5.48	5.17	4.42	2.23	2.18	[5.63]	[4.34]
5 (high)	10.31	9.74	7.44	6.82	6.90	3.20	2.53	[10.27]	[9.10]

State cut-offs (B/P spread): 0.04, 0.13, 0.24, 0.32 (GFKSW: −0.01, 0.12, 0.22, 0.27; Appendix Table 9). For every formation month, the strategy's monthly excess returns over the next h months are compounded and annualised; the table averages them by state. Windows end by December 2012. Bootstrap t: one-year mean divided by the standard deviation of a circular block bootstrap of monthly returns (24-month blocks, 999 draws) with the state calendar held fixed (baseline) and of its volatility-preserving variant (Section 3.3); it tests whether the state average differs from zero. Published GFKSW values in brackets.

the next-month L–H return has the opposite sign in states 2 to 4 (Appendix Table 9), and the one-year returns in states 3 and 4 are −1.6% and −3.0% against −6.84% and +3.36%. Because the middle states span narrow ranges of the spread, their classification is fragile: with calendar-year instead of June-lagged book equity 12% of the months change state, with value-weighted instead of equal-weighted B/P averages 50%. Yet the one-year L–H returns of states 2 to 4 move by at most 2.4 percentage points across these and three trimming variants, and no variant reproduces GFKSW's middle states (Appendix Table 12).

These estimates are imprecise. With the baseline bootstrap the one-year means are significant only in state 5 for L–H and in states 4 and 5 for BN, but this bootstrap over-rejects for the state-5 means when volatility follows its observed path, and with the volatility-preserving bootstrap the state-5 mean of L–H is not significant ($t = 1.47$); the bootstrap standard errors of the one-year L–H means are 5.1 to 6.2 percentage points. The contrast that carries GFKSW's argument, state 5 minus state 1, is 16.7 percentage points per year for L–H ($t = 1.99$, $p = 0.045$) and 9.4 points for BN ($t = 2.02$, $p = 0.035$). The cheap-minus-expensive contrasts are not robustly significant at the 5% level (Appendix Tables 18 and 19): across block lengths of 12 to 60 months the L–H t-statistic ranges from 1.96 to 2.14; the volatility-preserving bootstrap gives 1.74 (BN: 1.77); and under the observed volatility path a 5% test needs $|t|$ above 2.29 for L–H and 2.03 for BN. At three and five years the L–H contrast is 5.4 and 0.9 points per year and far from significant. The contrast also rests on one decade: state-5 months in 2000–2001 were followed by the collapse of the technology bubble (L–H: 63% in the following year), state-1 months in 2008–2009 by the rebound of high-beta stocks (−28%), and without the formation months of the 2000s the contrast falls to 2.4 points (Appendix Table 17).

A predictive regression avoids the classification step and tests a different hypothesis: a linear relation between the spread and the next-year return across all months, not a difference between the returns after the two extreme states. A one-standard-deviation higher spread predicts a 6.2 percentage points higher L–H return over the next 12 months ($t = 2.02$, Newey-West with 12 lags) and a 3.8 points higher BN return ($t = 3.30$). The spread is persistent (annual autocorrelation

0.53) and its innovations are negatively correlated with the returns, so on the 45 non-overlapping December observations we correct for the Stambaugh bias: the reduced-bias t-statistics are 1.47 for L–H and 2.95 for BN. Controlling for the market's excess return over the preceding year, which is positively correlated with the spread (0.33), lowers the spread's Newey-West t-statistics only slightly, from 2.02 and 3.30 to 1.79 and 3.09. In sample, the valuation signal therefore predicts the return of the beta-neutral portfolio significantly, whereas for the zero-cost portfolio it is not significant once the bias is corrected.

4.3 Value and momentum exposures

Table 3: Conditional alphas by initial B/P spread state, 1968–2012 (GFKSW Table 3)

	CAPM alpha				Four-factor alpha				Loadings
	12 months		60 months		12 months		60 months		12 months
State	Ours	GFKSW	Ours	GFKSW	Ours	GFKSW	Ours	GFKSW	HML / MOM
Panel A: low-minus-high (L–H)									
1 (low)	0.11	0.07	0.27	0.14	0.06	0.10	0.36	0.22	−0.04 / 0.07
	(0.30)	(0.10)	(0.73)	(1.15)	(0.21)	(0.31)	(1.31)	(1.30)	
2	0.55	0.45	0.31	0.36	0.15	0.18	0.21	0.15	0.49 / 0.14
	(1.72)	(0.90)	(1.04)	(1.23)	(0.59)	(0.66)	(0.95)	(0.72)	
3	0.22	−0.11	0.44	0.37	0.28	0.03	0.11	−0.11	0.36 / 0.11
	(0.74)	(−0.34)	(1.79)	(2.25)	(1.16)	(0.16)	(0.56)	(−0.67)	
4	0.35	0.46	0.38	0.39	0.00	0.06	−0.21	−0.09	0.70 / 0.24
	(1.18)	(2.62)	(1.34)	(2.69)	(0.00)	(0.24)	(−0.95)	(−0.58)	
5 (high)	0.68	1.04	0.78	0.92	0.12	0.40	0.06	0.29	0.72 / 0.28
	(2.01)	(5.11)	(2.77)	(4.28)	(0.45)	(1.86)	(0.30)	(1.72)	
Panel B: beta neutral (BN)									
1 (low)	0.02	0.00	0.10	0.02	0.02	0.03	0.18	0.11	0.00 / −0.02
	(0.10)	(−0.01)	(0.45)	(0.18)	(0.08)	(0.19)	(1.06)	(0.91)	
2	0.20	0.15	0.13	0.15	−0.06	0.04	0.08	0.06	0.23 / 0.06
	(1.06)	(0.44)	(0.73)	(0.67)	(−0.33)	(0.20)	(0.61)	(0.39)	
3	0.08	−0.09	0.24	0.21	0.12	−0.01	0.02	−0.07	0.21 / 0.08
	(0.44)	(−0.51)	(1.67)	(2.07)	(0.72)	(−0.07)	(0.15)	(−0.64)	
4	0.19	0.27	0.23	0.23	−0.06	0.06	−0.13	−0.06	0.41 / 0.17
	(1.09)	(2.56)	(1.36)	(2.91)	(−0.37)	(0.29)	(−0.98)	(−0.53)	
5 (high)	0.52	0.68	0.44	0.52	0.30	0.34	0.06	0.16	0.35 / 0.18
	(2.55)	(5.09)	(2.62)	(4.07)	(1.65)	(1.85)	(0.43)	(1.02)	

Alphas in percent per month. For every formation month the strategy's excess returns over the next 12 or 60 months are regressed on the market or on MKT, SMB, HML and MOM, and the intercepts are averaged by state. In parentheses: for our estimates the bootstrap t-statistic (as in Table 2); for GFKSW their published Newey-West t-statistic. Our GFKSW-style statistics are in Appendix Table 16. HML / MOM: average four-factor loadings over 12-month windows.

Table 3 replicates GFKSW's attribution. In state 5 the L–H portfolio has a CAPM alpha of 0.68% per month over the following year against GFKSW's 1.04%, and its GFKSW-style t-statistic is 2.00 against their 5.11, so even our reading of their procedure does not reproduce their precision. Adding SMB, HML and MOM lowers the alpha to 0.12% (GFKSW: 0.40%), while the HML loading of L–H rises from −0.04 in state 1 to 0.72 in state 5 and the MOM loading from 0.07 to 0.28, the pattern of GFKSW's Table 4 (Appendix Table 13; rolling correlations in Appendix Figure 4). The loadings show a value exposure of the cheap-state premium, not a full explanation: when low-beta stocks are cheap relative to high-beta stocks, a portfolio long low-beta and short high-beta stocks is long value.

With the baseline bootstrap, no four-factor alpha is significant at any of the five horizons from 12 to 60 months (largest $|t| = 1.88$), whereas the state-5 CAPM alphas are significant at every horizon for both portfolios with this bootstrap, which over-rejects here. The volatility-preserving bootstrap keeps them significant only at 48 and 60 months for L–H and 24, 48 and 60 months for BN, not over the first year (L–H: $t = 1.32$). The state-5 CAPM alpha of L–H also comes largely from windows that start in the 1980s: without them it falls from 0.68% to 0.07% per month.

With book equity from Davis et al. (2000), the 1930–2012 sample shows the same ordering of the extreme states at every horizon, but after the cheapest state L–H gains only in the first two years (Appendix Table 20).

5 Post-sample performance and valuation timing

Table 4: Performance in GFKSW's sample, after its end and after publication

Portfolio	Excess (%)	t	Sharpe	Beta	CAPM α (%)	t	4F α (%)	t
Panel A: GFKSW sample, 1968–2012 (540 months)								
Low beta (Q1)	5.4	2.74	0.43	0.63	2.1	1.77	0.4	0.42
High beta (Q5)	3.5	0.88	0.14	1.45	−4.0	−2.73	−1.0	−0.75
Low – high	1.9	0.58	0.09	−0.82	6.2	2.48	1.4	0.67
Beta neutral (25%)	4.5	2.87	0.45	0.27	3.1	2.11	0.7	0.54
Beta neutral (ex ante)	4.2	2.68	0.44	0.23	3.0	2.02	1.0	0.82
Market	5.2	2.06	0.33	1.00	–		–	
Panel B: after the original sample end, 2013–2024 (144 months)								
Low beta (Q1)	9.5	3.91	0.84	0.59	1.7	0.84	−0.3	−0.13
High beta (Q5)	16.8	3.09	0.74	1.39	−1.7	−0.65	1.1	0.44
Low – high	−7.3	−1.46	−0.38	−0.80	3.4	0.87	−1.4	−0.34
Beta neutral (25%)	5.3	2.33	0.55	0.24	2.1	0.89	−0.5	−0.23
Beta neutral (ex ante)	5.1	2.17	0.52	0.22	2.1	0.88	−0.6	−0.25
Market	13.3	3.87	0.89	1.00	–		–	
Panel C: after publication, 2016–2024 (108 months)								
Low beta (Q1)	8.0	2.81	0.67	0.57	0.6	0.26	−0.9	−0.42
High beta (Q5)	19.0	2.81	0.76	1.41	0.7	0.24	3.1	0.98
Low – high	−11.0	−1.81	−0.52	−0.84	−0.1	−0.03	−4.0	−0.83
Beta neutral (25%)	3.2	1.23	0.32	0.21	0.4	0.16	−1.7	−0.62
Beta neutral (ex ante)	2.8	1.04	0.27	0.19	0.3	0.11	−1.9	−0.66
Market	13.0	3.01	0.80	1.00	–		–	

Annualised mean excess returns, Sharpe ratios, realised market betas and annualised CAPM and four-factor alphas. Newey-West t-statistics with $\lfloor 4(T/100)^{2/9} \rfloor$ lags (5 in Panel A, 4 in Panel B, 4 in Panel C).

GFKSW's sample ends in December 2012 and their article appeared in mid-2015, so 2013–2024 lies after the original sample end, and only 2016–2024 after publication. Neither period is an untouched out-of-sample test for us, since the data existed before we specified anything. Classified with the 1968–2012 breakpoints, 106 of the 144 post-sample formation months fall into state 1, 36 into state 2, 2 into state 3 and none into states 4 or 5 (after publication: 72, 34 and 2). The period therefore cannot test GFKSW's central prediction, that low-beta stocks win after starting cheap.

What it does show is mostly market beta (Table 4). The annualised mean excess return of L–H over T-bills was −7.3% in 2013–2024 and −11.0% in 2016–2024. In a CAPM decomposition for 2013–2024, its realised beta of −0.80 times the market's mean excess return of 13.3% gives −10.7% per year, against a CAPM alpha of 3.4% ($t = 0.87$): the decomposition attributes the negative excess return mainly to the short market exposure in a rising market. Of the beta-neutral portfolio's mean excess return of 5.3% per year, the CAPM attributes 3.2 points to its beta of 0.24; the shrunk-beta version, with a realised beta of −0.06, earned 1.7% ($t = 0.61$, Appendix Table 23). The low-beta quintile had a lower Sharpe ratio than the market (0.84 against 0.89) but a positive CAPM alpha of 1.7% per year ($t = 0.84$); twelve years cannot distinguish a premium of the 1968–2012 size (2.1% per year) from zero. With only 12 non-overlapping years, the predictive regression after 2012 has little power: the reduced-bias L–H slope is of similar size as before (4.3 against 4.9 percentage points per standard deviation, $t = 0.45$), whereas the BN slope turns negative ($t = -0.97$).

Can the valuation signal be used in real time? With breakpoints estimated only from earlier months, the one-year L–H return after a real-time state-5 signal is 11.6% and after a state-1 signal

−6.5%. GFKSW explicitly do not propose a timing rule; Table 5 tests the simplest one their results suggest: hold the long-short portfolio only when the real-time state is 4 or 5, in a second variant only if the strategy's own trailing 12-month return is also positive (time-series momentum of the strategy, not the MOM factor). For L–H, timing turns a negative net Sharpe ratio into a small positive one, mainly by avoiding the short market exposure, but not significantly (Holm $p = 0.57$); for the beta-neutral portfolio it lowers the net Sharpe ratio from 0.48 to 0.36. After 2012 the real-time states never exceeded 3, so the rule held T-bills throughout. The tested rules bring no statistically significant net Sharpe ratio improvement, which does not prove that timing is useless; timing factors with their own valuation spreads is known to be hard (Asness et al. 2017).

Table 5: Real-time valuation timing of the long-short portfolios, 1978–2024 (exploratory)

Rule	Invested (%)	Excess (%)	SR	CAPM α (%)	4F α (%)	SR net	ΔSR net	Holm p
Low-minus-high (L–H)								
Always invested	100	−1.9	−0.09	5.9	1.9	−0.12	–	–
		(−0.58)		(2.39)	(0.90)			
Valuation (states 4–5)	38	1.4	0.10	4.3	2.9	0.07	0.19	0.57
		(0.64)		(1.87)	(1.76)		(0.13)	
Valuation and own momentum	15	1.1	0.13	2.2	0.0	0.10	0.22	0.57
		(1.12)		(1.87)	(−0.01)		(0.16)	
Beta neutral (BN)								
Always invested	100	5.0	0.50	3.2	0.9	0.48	–	–
		(3.58)		(2.18)	(0.78)			
Valuation (states 4–5)	38	2.6	0.39	2.1	1.0	0.36	−0.12	0.69
		(2.57)		(1.89)	(1.02)		(0.12)	
Valuation and own momentum	27	2.0	0.39	1.5	0.3	0.35	−0.13	0.69
		(2.46)		(1.79)	(0.33)		(0.14)	

States are classified each month with quintile breakpoints from earlier formation months only (expanding window, at least ten years from 1968). The portfolio is held when the real-time state is 4 or 5 (valuation), additionally only when its own trailing 12-month return is positive (valuation and own momentum), and otherwise T-bills are held. Annualised; in parentheses Newey-West t-statistics and, for ΔSR net, the Ledoit-Wolf standard error. SR net deducts 10 bp per dollar traded, the lending fee and the cost of switching in and out. ΔSR net: difference to the always-invested net Sharpe ratio; Holm-adjusted p-values over the four comparisons.

6 Beta forecasts and portfolio performance

GFKSW sort on betas from 60 monthly returns. Daily returns contain more information about co-movement but are noisier and suffer from non-synchronous trading (Dimson 1979; Hollstein et al. 2019). Appendix Table 21 compares seven estimators, which differ in return frequency and window. Holding the window at 36 months, daily returns forecast the realised beta of the same portfolios significantly better than monthly returns (Diebold-Mariano $t = -4.23$). Holding the frequency fixed, short windows are worse: the 36-month monthly beta is less accurate than the 60-month one ($t = 3.63$), and the three-month daily beta less accurate than the 36-month one ($t = 4.08$). Against GFKSW's 60-month monthly beta, only the 36-month daily beta and the Dimson-corrected 12-month daily beta have significantly smaller squared errors (Holm $p = 0.014$ and 0.002), and these results survive a Holm adjustment over all ten forecast comparisons (largest adjusted $p = 0.021$). Daily betas are thus better risk forecasts when the window is long enough, in line with Hollstein et al. (2019).

As strategies, the estimated ranking reverses. The ex-ante beta-neutral portfolio formed on the 60-month monthly beta has the highest gross Sharpe ratio (0.45); every daily specification has a lower one (0.30 to 0.37), but no difference in gross Sharpe ratios is significant after the Holm adjustment (smallest adjusted $p = 0.73$). The valuation effect is also largest with the 60-month monthly beta. A plausible but untested explanation is that slow-moving monthly betas capture persistent firm characteristics such as value and leverage, whereas daily betas measure current co-movement. The daily-beta portfolio of Section 7 uses the 12-month daily beta, which is not among the significantly better forecasters; we do not switch to the best forecaster after the fact.

7 Long-only implementation after trading costs

Where does the premium come from? In 1968–2012 the high-beta quintile's CAPM alpha was −4.0% per year ($t = -2.73$) and the low-beta quintile's 2.1% ($t = 1.77$, Table 4): 65% of the L–H alpha comes from the short side. Most investors cannot or will not short and are judged against a cap-weighted benchmark (Baker et al. 2011; Blitz et al. 2014); for them the question is whether a long-only low-beta portfolio beats the portfolio they would otherwise hold. All portfolios in Table 6 are formed from the same large-cap universe over 1968–2024, rebalanced at month ends and charged costs on their own turnover: the value-weighted low-beta quintile; a buffer rule that buys stocks entering the lowest 20% of betas and sells them only when they leave the lowest 30%; quarterly rebalancing; the quintile with target weights capped at 5% per stock at each rebalancing; the quintile formed on 12-month daily betas; and two long-only global minimum-variance portfolios with a 3% weight cap, one estimated with the constant-correlation shrinkage of Ledoit and Wolf (2004) on 60 monthly returns, the other with the nonlinear shrinkage of Ledoit and Wolf (2020) on the daily returns of the last 12 months. With on average 517 (monthly) and 550 (daily) eligible stocks, the sample covariance matrix is singular in both cases. The benchmarks are the cap-weighted portfolio of the same universe and the equally weighted $1/N$ portfolio (DeMiguel et al. 2009).

Table 6: Long-only low-beta and minimum-variance portfolios versus the cap-weighted benchmark, 1968–2024

	Low-beta quintile					Minimum variance		Benchmarks	
	Q1	Buffer	Quarterly	Capped	Daily β	Monthly	Daily	$1/N$	Cap-wt.
Panel A: gross performance									
Excess return (%)	6.3	6.5	6.2	6.6	5.9	6.6	6.1	7.6	6.9
Volatility (%)	12.3	12.3	12.3	12.4	11.8	12.1	11.5	16.7	15.3
Sharpe ratio	0.51	0.53	0.51	0.53	0.49	0.54	0.53	0.45	0.45
Max. drawdown (%)	42	40	41	42	40	41	40	53	50
Market beta	0.62	0.65	0.62	0.63	0.56	0.55	0.57	1.03	0.96
CAPM α (%)	2.0	2.0	1.9	2.2	1.9	2.8	2.2	0.4	0.2
	(1.85)	(2.07)	(1.86)	(2.01)	(1.84)	(2.34)	(2.17)	(0.75)	(0.80)
Four-factor α (%)	0.5	0.6	0.4	0.6	0.6	1.2	0.8	0.2	0.2
	(0.52)	(0.75)	(0.47)	(0.67)	(0.54)	(1.13)	(0.83)	(0.37)	(1.54)
HML loading	0.20	0.17	0.20	0.20	0.25	0.30	0.25	0.12	0.00
Active return (%)	−0.6	−0.4	−0.6	−0.3	−1.0	−0.2	−0.7	0.7	–
	(−0.47)	(−0.34)	(−0.55)	(−0.24)	(−0.78)	(−0.17)	(−0.58)	(1.18)	
Tracking error (%)	8.6	7.8	8.6	8.6	9.9	10.4	9.7	4.2	–
Information ratio	−0.07	−0.05	−0.07	−0.03	−0.10	−0.02	−0.08	0.17	–
Panel B: costs (scenario: 10 bp per dollar traded)									
Turnover (% p.a.)	179	70	106	182	291	243	432	126	24
Sharpe ratio, net	0.50	0.52	0.50	0.52	0.47	0.52	0.49	0.45	0.45
ΔSR net	0.05	0.08	0.05	0.07	0.02	0.08	0.05	0.00	–
	[0.08]	[0.08]	[0.08]	[0.09]	[0.10]	[0.10]	[0.09]	[0.04]	
p-value	0.53	0.32	0.52	0.41	0.80	0.44	0.60	0.99	–
Holm p-value	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	–
Break-even cost (bp)	50	192	86	63	20	53	23	10	–
Panel C: diversification (averages over rebalancing dates)									
Positions	110	134	110	110	110	47	82	553	553
Largest weight (%)	12.3	10.4	12.3	5.0	11.7	3.0	3.0	0.2	4.5
Effective N	29	34	29	44	38	38	54	553	119

Annualised; Newey-West t-statistics (6 lags) in parentheses, standard errors in brackets. Capped: quintile weights capped at 5%; Daily β: quintile formed on 12-month daily betas; minimum variance with a 3% cap, covariance matrix from 60 monthly returns (constant-correlation shrinkage) or from the daily returns of the last 12 months (nonlinear shrinkage). Cap-wt.: value-weighted portfolio of all universe stocks (benchmark of the tests); the Fama-French market has an excess return of 6.9% and a Sharpe ratio of 0.44. Turnover: purchases plus sales relative to drifted weights. ΔSR net: Ledoit-Wolf (2008) HAC test against the cost-bearing benchmark; the low-beta quintile is the primary comparison (unadjusted p), Holm adjustment over the 8 candidates. Break-even: one-way cost at which the net Sharpe ratio equals the benchmark's (values above 50 bp extrapolate beyond the highest cost scenario). Effective N: $1/\sum w_i^2$. Liquidity: Appendix Table 22.

The low-beta quintile delivers lower risk: 12.3% volatility, a beta of 0.62 and a maximum drawdown

of 42%, against 15.3%, 0.96 and 50% for the benchmark, at a lower excess return (6.3% against 6.9% per year). Its Sharpe ratio is 0.51 against 0.45 gross and 0.50 against 0.45 after the base-case cost. This primary comparison is inconclusive: the difference of 0.05 has a standard error of 0.08 ($p = 0.53$), consistent both with no advantage and with a sizeable one. Judged as an active manager, the portfolio trails the benchmark by 0.6% per year with a tracking error of 8.6% (information ratio −0.07): exactly the profile that makes benchmarked managers avoid low-beta stocks (Baker et al. 2011). Its CAPM alpha of 2.0% per year ($t = 1.85$) shrinks to 0.5% with the four-factor model (95% interval −1.3% to 2.2%), alongside an HML loading of 0.20.

The low-beta quintile turns over 179% of its value per year, against 24% for the benchmark, and its net Sharpe ratio would equal the benchmark's at a one-way cost of 50 basis points (Appendix Figure 6). The buffer rule cuts turnover to 70% and raises the break-even cost to 192 basis points, an extrapolation beyond the tested range of 0 to 50 basis points. In sample, the 5% cap raises the effective number of positions from 29 to 44 and turnover from 179% to 182%. Under the assumed cost model, the estimated net Sharpe ratio rises from 0.50 to 0.52. These are descriptive results. The daily nonlinear-shrinkage implementation has the lowest realized volatility among the portfolios considered (11.5%). Since return frequency, estimation window, estimator and stock selection differ at the same time from the monthly version, this is no isolated evidence that the estimator is superior. Its weights change so quickly that it turns over 432% per year and its net Sharpe ratio falls to 0.49 (break-even 23 basis points). The monthly version has the largest CAPM alpha (2.8%, $t = 2.34$) and the highest net Sharpe ratio (0.525), closely followed by the buffer rule (0.521) and the capped quintile (0.515). No net Sharpe ratio of the 8 candidates differs significantly from the benchmark's after the Holm adjustment, and since 2013 every one of them has had a lower net Sharpe ratio than the benchmark (low-beta quintile: 0.83 against 0.94).

The low-beta quintile is also concentrated: its largest position averages 12.3% of the portfolio and its effective number of stocks is 29 against 119 for the benchmark. Its holdings rank like the benchmark's by dollar volume (Appendix Table 22). Proportional costs are modelled as scenarios, but actual execution costs and capacity, which depend on trade size and market impact, are not estimated. For investors who can short (Appendix Table 23), the beta-neutral portfolio keeps a net Sharpe ratio of 0.44 over 1968–2024, the nearly beta-neutral version with shrunk betas only 0.26, and L–H a negative one despite a gross CAPM alpha with $t = 2.58$; no long-short four-factor alpha is significant before or after costs.

8 Risk measurement and performance across holding periods

Month-end data hide intra-month risk, and a strategy is experienced through the windows over which investors judge it, typically three to five years. We rebuild every portfolio's daily return path from CRSP's daily file with unchanged holdings, monthly rebalancing and weights drifting within the month; long-short positions are also reset only at month ends, and all returns are gross of costs. Every stock-month of the cap-weighted universe portfolio compounds to its monthly return except 192 delisting months of 378,535; their treatment changes the volatilities in Table 7 by at most 0.01 percentage points and the Sharpe ratios by at most 0.002 (other statistics: Appendix Table 11), so the comparison isolates the measurement frequency.

Table 7 shows that frequency matters for risk measurement more than for the ranking. Annualised daily volatility exceeds monthly volatility for the low-beta quintile (13.3% versus 12.3%) and the cap-weighted benchmark (16.8% versus 15.3%), but not for the minimum-variance portfolio. With serially uncorrelated daily returns the two would agree up to sampling error and compounding effects. The variance ratio of overlapping 21-day to daily log returns (Lo and MacKinlay 1988),

Table 7: The same portfolios measured with daily and with monthly returns, 1968–2024

	Volatility (%)		Sharpe ratio		Max. drawdown (%)		Excess kurtosis		Market beta	
Portfolio	Daily	Monthly	Daily	Monthly	Daily	Monthly	Daily	Monthly	Daily	Monthly
Low beta (Q1)	13.3	12.3	0.48	0.51	46	42	25.3	1.3	0.67	0.62
High beta (Q5)	25.6	24.6	0.26	0.26	86	82	9.9	1.3	1.44	1.44
Min. variance (monthly)	11.8	12.1	0.56	0.55	43	41	26.6	1.7	0.58	0.55
Cap weight	16.8	15.3	0.42	0.45	54	50	16.6	1.4	1.01	0.96
Beta neutral (25%)	10.1	10.1	0.47	0.47	32	31	19.8	1.7	0.32	0.26
Low – high	19.2	20.2	−0.01	0.00	88	87	11.9	3.2	−0.72	−0.82
Market	16.6	15.8	0.42	0.44	55	50	14.8	1.6	1.00	1.00

Volatility and Sharpe ratios are annualised with 252 days or 12 months; drawdowns, kurtosis and betas are not. Monthly returns compound the daily returns of the same holdings, so rebalancing is monthly in both cases. Kurtosis: excess kurtosis of daily or monthly excess returns, i.e. different horizons. Beta: slope on the market excess return at each frequency. Gross returns; daily paths from CRSP's daily file (dsf_v2); excess returns over the T-bill return compounded from daily rates. Long-short portfolios: total returns, legs reset at month ends.

which differs from the squared ratio of the volatilities in the table because it uses log returns and overlapping windows rather than calendar months, is 0.896 for the low-beta quintile and 0.904 for the benchmark, with heteroskedasticity-robust z-statistics of −0.98 and −1.02: the gap is consistent with mild negative serial correlation within the month, but the data do not establish it. Daily drawdowns are at least as deep, because intra-month troughs are invisible in month-end data, and daily returns are far more fat-tailed. The low-beta quintile's beta is higher with daily data (0.67 versus 0.62), and the beta-neutral portfolio's daily beta of 0.32 (Newey-West standard error 0.02) confirms that it carries market risk at both frequencies. Sharpe ratios change by at most 0.03 between frequencies. The monthly columns differ slightly from Table 6 only through the T-bill return compounded from daily rates (up to 11 basis points per month away from the monthly series) and, for the market, compound the daily Fama-French factor; with the monthly T-bill series the Sharpe ratios of Table 6 are reproduced exactly.

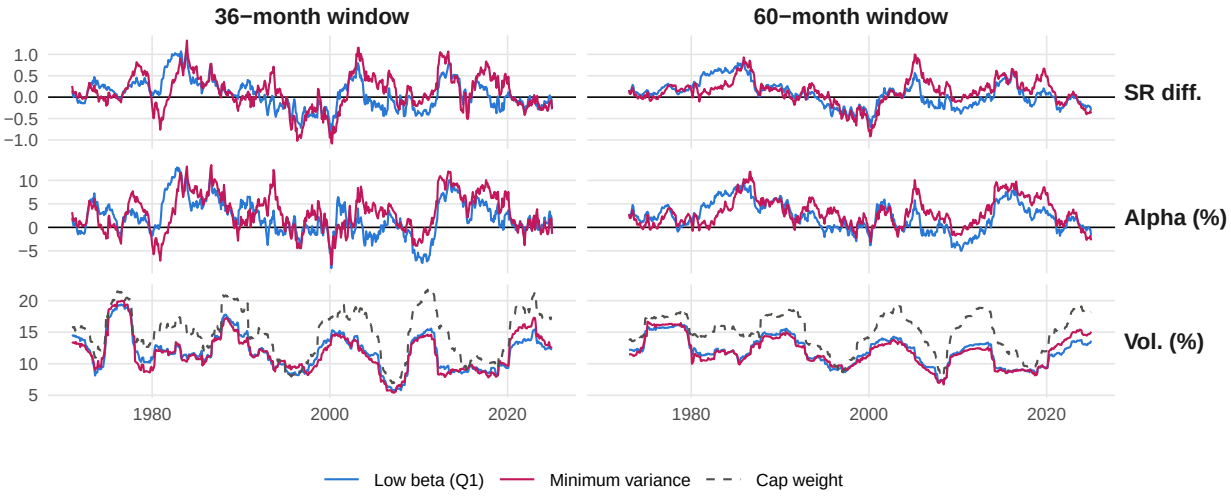


Figure 3: Rolling 36- and 60-month evaluation windows of monthly gross returns, plotted at the window's last month; the first windows end in December 1970 (36 months) and December 1972 (60 months), the last in December 2024. Top: annualised Sharpe ratio (from excess returns) of the low-beta quintile and of the minimum-variance portfolio (monthly estimation) minus that of the cap-weighted benchmark of the same universe. Middle: CAPM alpha of the low-beta quintile and of the minimum-variance portfolio (monthly estimation), in percent per year. Bottom: annualised volatility, in percent, of the low-beta quintile, the minimum-variance portfolio and the cap-weighted benchmark.

Rolling evaluation windows (Figure 3), distinct from the fixed 60-month estimation windows of betas and covariances, show what an investor would have experienced. The risk reduction is the most stable property of the low-beta quintile: its volatility is below the benchmark's in 97% of

36-month and 100% of 60-month windows, at 0.58 to 1.15 times the benchmark's volatility over 36 months. Its return advantage is not: it has a higher Sharpe ratio than the benchmark in 55% of 36-month and 60% of 60-month windows, a positive CAPM alpha in 75% of 60-month windows, and a higher raw return than the benchmark in only 44% of 60-month windows. The minimum-variance portfolio has the higher Sharpe ratio in 72% of 60-month windows. Overlapping windows are strongly dependent, so these shares describe experience, not independent evidence. The pattern is consistent with the benchmarking explanation of why the anomaly persists (Baker et al. 2011): a strategy that trails its benchmark for years is hard to hold, even when its risk-adjusted return is higher over the full sample.

9 Conclusion

GFKSW's unconditional results replicate almost exactly. The one-year conditional L–H returns also reproduce the pattern at the extremes: positive after the cheapest state and negative after the most expensive state. The middle states do not replicate, and, in our reconstruction and simulations, Newey-West statistics of averages over overlapping windows overstate precision. The cheap-minus-expensive contrasts are not robustly significant at the 5% level. That of the zero-cost portfolio rests largely on the technology bust and the 2009 rebound. A bias-corrected predictive regression, which tests a different hypothesis, shows in sample that valuation predicts the beta-neutral portfolio's return, but the conditional premium coincides largely with a value exposure, no four-factor alpha is significant with the block bootstraps or calendar-time regressions, and the tested timing rules did not significantly improve the net Sharpe ratio. Since 2013 valuations have never been cheap enough to test GFKSW's central prediction, and a CAPM decomposition attributes the zero-cost portfolio's negative excess returns mainly to its negative beta in a rising market.

For long-only investors, an implementation gap remains: about two thirds of the L–H portfolio's CAPM alpha in 1968–2012 comes from shorting high-beta stocks. The most stable result is the historical risk reduction. A long-only low-beta portfolio cut volatility by about a fifth and the maximum drawdown from 50% to 42%, but neither a net Sharpe ratio advantage over the benchmark nor a timing advantage is established, and investors must accept deep drawdowns and long benchmark underperformance. In the variants studied, the buffer rule reduces turnover and the weight cap concentration, without proven general superiority. Better risk forecasts from daily betas did not yield significantly higher gross Sharpe ratios, and the daily nonlinear-shrinkage portfolio has the lowest realized volatility but no higher net Sharpe ratio than the low-beta quintile.

Four limitations remain: the valuation states rest on few episodes, so test sizes depend on how their volatility is modelled; CRSP data vintages differ from GFKSW's; the proportional cost model uses scenario values and ignores market impact, financing spreads and borrowing constraints; and the analysis is limited to US large caps.

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Appendix

A. Replication design, data treatment and sensitivity

Table 8: Replication design

Item	GFKSW (2015)	This paper
Stock data	CRSP 1925–2012 (legacy format), share codes 10/11	CRSP CIZ format (Tidy Finance database), US common stocks; analysis ends Dec. 2024
Delisting returns	Performance-related delistings: −30% (NYSE/AMEX), −55% (NASDAQ)	CRSP monthly delisting-month returns; GFKSW rule as sensitivity (Table 10); daily paths with delisting-day records
Market index	CRSP value-weighted	Fama–French market total return (MKT–RF + RF; same mean as panel VW return)
Universe	Top third by NYSE breakpoints, USD 2–1,000	Same; stocks delisted in the formation month excluded; universe restricted to December-FYE firms only in Table 14
Beta	60 months (min. 24), monthly	Identical; also 36/120 months and daily 3/12/36 months
Book equity	Compustat, December FYE, two years; French data before 1951	Identical; Davis-Fama-French book equity before Compustat coverage
B/P timing	Prior fiscal year; market value at month end	December book equity from June of the following year (calendar-year timing as sensitivity)
B/P outliers	Top and bottom 0.5% of B/P deleted; 108 months per state	Stock-level B/P trimmed cross-sectionally each month; 108 months per state
Long-short excess returns	L–H: $r_{Q1} - r_{Q5}$; BN: $r_{Q1} - 0.25r_{Q5} - 0.75r_f$; three-month T-bills	Same definitions; one-month T-bills
Beta-neutral ratio	Fixed 25% (full-sample average betas)	Fixed 25%; ex-ante and shrunk-beta ratios as extensions
Multi-period returns	Annualised future cumulative excess return; monthly rebalancing	Chained monthly excess returns, annualised; windows end by Dec. 2012
Inference	OLS t; Newey-West for overlapping windows	Replicated statistics plus circular block bootstrap for window averages (block-length, volatility and size checks); Newey-West elsewhere

Table 9: Next-month returns on beta portfolios by initial B/P spread state, 1968–2012 (replication of GFKSW Table 2, Panel A)

			Beta quintile (%)					Long-short (%)			
State	B/P spread	N	Q1	Q2	Q3	Q4	Q5	L–H	GFKSW	BN	GFKSW
1 (low)	−0.66 to 0.04 [−0.55 to −0.01]	101 [104]	0.22	0.37	0.40	0.32	0.33	−0.10	−0.21	−0.07	−0.12
								(−0.17)		(−0.26)	
2	0.04 to 0.13 [−0.01 to 0.12]	104 [101]	0.78	1.05	1.09	1.01	0.92	−0.14	0.21	0.31	0.69
								(−0.35)		(1.45)	
3	0.13 to 0.24 [0.12 to 0.22]	105 [108]	0.86	0.95	0.90	0.67	0.65	0.21	−0.40	0.38	0.21
								(0.43)		(1.37)	
4	0.24 to 0.32 [0.22 to 0.27]	114 [106]	1.31	1.54	1.53	1.91	1.74	−0.43	0.23	0.50	0.47
								(−0.95)		(1.91)	
5 (high)	0.32 to 0.76 [0.27 to 0.59]	116 [94]	1.26	0.80	0.42	0.36	0.02	1.24	0.87	0.77	0.80
								(1.61)		(2.22)	

States are quintiles of the 540 monthly B/P spreads (average B/P of the low-beta minus the high-beta quintile, stock-level B/P trimmed at 0.5% in each tail every month); each state contains 108 months. Value-weighted returns in percent in the month after formation; ordinary t-statistics in parentheses, as in GFKSW. N: average number of stocks per extreme quintile. BN returns are in excess of the T-bill rate. GFKSW: their Table 2, Panel A; their B/P spread ranges and stock counts in brackets.

Table 10: Treatment of delisting months and its effect, 1968–2012

1968–2012	Baseline	GFKSW rule	Delisting months dropped	Formation-month delistings kept	Daily-file return
Held stock-months affected	–	6	776	165	4
Mean return Q1 (% per month)	0.885	0.885	0.880	0.884	0.885
Mean return Q5 (% per month)	0.729	0.729	0.724	0.727	0.729
Mean return L–H (% per month)	0.156	0.156	0.156	0.157	0.156
Mean excess return BN (% per month)	0.378	0.378	0.374	0.377	0.378
Maximum drawdown Q1 (%)	42.2	42.2	42.5	42.2	42.2
Maximum drawdown Q5 (%)	81.6	81.6	81.6	81.6	81.6
L–H, one year, state 1 (% p.a.)	−4.93	−4.93	−4.97	−4.92	−4.93
L–H, one year, state 5 (% p.a.)	11.79	11.79	11.88	11.80	11.79

Baseline: holding returns from CRSP's monthly file, including delisting months; stocks whose last trading day falls in the formation month are not bought. GFKSW rule: in held months in which the stock is dropped by its exchange (CIZ delisting action GDR, the legacy 500 codes), the return after the last trading day is set to −30% (NYSE, AMEX) or −55% (NASDAQ); 25 held stock-months over 1929–2024, 6 in 1968–2012. Delisting months dropped: held stocks without a regular month-end record removed and weights renormalised. Formation-month delistings kept: stocks delisted in the formation month bought, their proceeds held at a zero return. Daily-file return: in the delisting months in which CRSP's monthly return excludes the delisting-day return of its daily file, the compounded daily return is used. States as in Table 9. Over 1930–2024, 20 large-cap stock-months without trades (all before 1968) are valued at the last trade price.

Table 11: Sensitivity of the daily statistics to the treatment of delisting months, 1968–2024

Portfolio	Volatility (pp)	Sharpe ratio	Excess kurtosis	Max. drawdown (pp)	Market beta
Low beta (Q1)	0.001	0.0018	0.012	0.046	0.00008
High beta (Q5)	0.002	0.0008	0.001	0.042	0.00009
Minimum variance	0.001	0.0014	0.013	0.037	0.00002
Cap weight	0.001	0.0010	0.004	0.024	0.00006
Equal weight	0.002	0.0017	0.007	0.031	0.00010
Beta neutral (25%)	0.001	0.0019	0.011	0.037	0.00006
Low – high	0.002	0.0002	0.001	0.036	0.00000

Largest absolute difference from the baseline daily reconstruction (Section 8) across three alternatives: (i) in the 4 stock-months in which CRSP's monthly return excludes an in-month delisting-day return of its daily file (dsf_v2), the daily file's return is used; (ii) the earlier reconstruction from the database's daily table (crsp_daily, without delisting-day records), with the whole difference to the monthly return booked on the last trading day; (iii) daily returns without any delisting return. Baseline: daily file; delisting-day records dated in the first days of the next month (188 stock-months of the cap-weighted portfolio), which CRSP's monthly file credits to the held month, booked on the held month's last trading day; in the 4 conflicting months the monthly file is followed. Days without a record earn zero (proceeds held in cash until the month-end rebalancing).

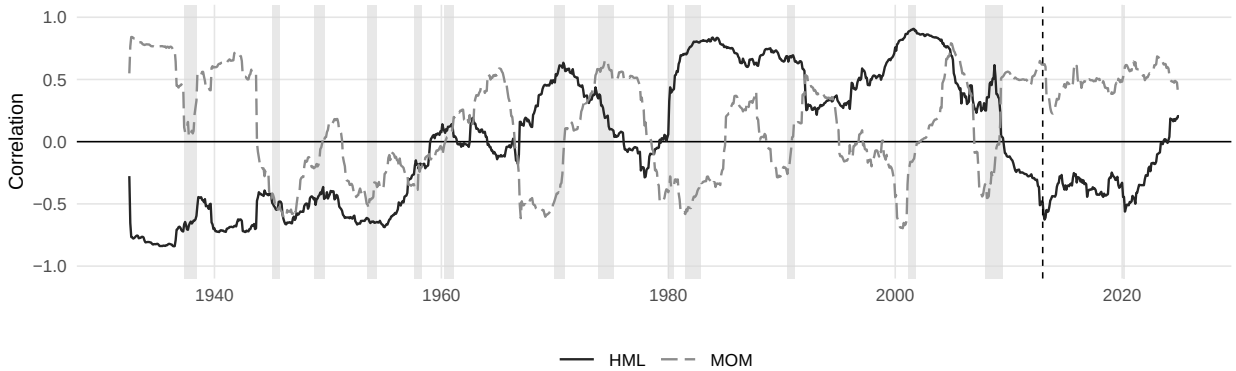


Figure 4: Rolling 48-month correlations between the low-minus-high portfolio and the HML and MOM factors, windows ending between June 1932 and December 2024 (GFKSW Figure 4, extended), plotted at the window's last month. Shaded areas are NBER recessions; the dashed line marks December 2012.

Table 12: Sensitivity of the conditional returns to the state classification, and three multi-period return concepts, 1968–2012

		Valuation state				
Specification	Same state (%)	1 (low)	2	3	4	5 (high)
L–H, one-year return (%)						
Stock-level B/P, cross-sectional trim (baseline)	100	−4.9	7.3	−1.6	−3.0	11.8
Stock-level B/P, pooled 1968–2012 trim	88	−4.9	7.6	−2.5	−1.0	10.3
Untrimmed B/P, 0.5% of months trimmed	82	−3.8	4.9	0.2	−1.4	10.3
Untrimmed B/P, no trimming	82	−5.3	5.0	0.1	−1.4	11.2
Value-weighted B/P within the quintiles	50	−5.4	5.4	−2.8	−1.5	13.8
Book equity of calendar year $y-1$ in all months of y	88	−4.7	7.4	−3.3	−1.5	11.7
BN, one-year return (%)						
Stock-level B/P, cross-sectional trim (baseline)	100	0.9	1.7	4.2	6.2	10.3
Stock-level B/P, pooled 1968–2012 trim	88	1.5	0.9	4.3	5.8	10.9
Untrimmed B/P, 0.5% of months trimmed	82	0.9	0.5	4.9	6.2	10.7
Untrimmed B/P, no trimming	82	0.7	0.6	4.9	6.4	10.7
Value-weighted B/P within the quintiles	50	0.9	5.3	3.4	2.8	10.9
Book equity of calendar year $y-1$ in all months of y	88	0.7	1.7	3.9	6.8	10.2
L–H over five years, baseline states (% per year; last row: % of initial capital)						
Chained excess return (baseline)		1.6	1.3	−1.1	−3.7	2.4
Relative wealth $\prod(1+r_{Q1})/\prod(1+r_{Q5})$		4.8	5.2	1.5	2.5	8.6
Wealth difference $\prod(1+r_{Q1})-\prod(1+r_{Q5})$ (cumulative)		22.7	25.4	4.0	0.0	46.6

Same state: share of months classified in the same state as in the baseline. Calendar-year book equity is used from January, up to five months before the assumed June availability date, and serves only as a check on GFKSW's "prior fiscal year". The three return concepts describe different investments (monthly rebalanced long-short position, two separate buy-and-hold portfolios, a long-short position never rebalanced) and are not alternative estimates of one quantity.

Table 13: Average HML and MOM loadings of the low-minus-high portfolio by B/P spread state, 1968–2012 (GFKSW Table 4)

	HML loading				MOM loading			
State	$t+12$	GFKSW	$t+60$	GFKSW	$t+12$	GFKSW	$t+60$	GFKSW
1 (low)	−0.04	[0.11]	0.11	[0.31]	0.07	[0.18]	0.02	[0.03]
2	0.49	[0.33]	0.46	[0.52]	0.14	[0.08]	0.16	[0.14]
3	0.36	[0.73]	0.58	[0.79]	0.11	[0.15]	0.12	[0.17]
4	0.70	[0.65]	0.81	[0.71]	0.24	[0.23]	0.13	[0.12]
5 (high)	0.72	[0.53]	0.79	[0.65]	0.28	[0.24]	0.17	[0.19]

Average four-factor slopes over overlapping windows, estimated as in Table 3. GFKSW values in brackets.

Table 14: Table 1 restricted to large caps with a December fiscal year-end book value, 1968–2012

Beta quintile	Arithmetic	Geometric	Beta	Sharpe	t-statistic
1 (low)	0.88	0.82	0.45	0.43	5.63
2	0.89	0.80	0.73	0.38	4.95
3	0.94	0.83	0.95	0.38	4.68
4	0.78	0.64	1.19	0.23	3.42
5 (high)	0.70	0.46	1.69	0.13	2.37
Low – high	0.18	0.03	−1.24	0.12	0.78
Beta neutral	0.71	0.67	0.03	0.46	5.61
Market	0.87	0.76	1.00	0.33	4.36

Same construction as Table 1, but the universe additionally requires a valid December-FYE book-to-price ratio. Returns in percent per month.

Table 15: Alternative estimator: calendar-time regressions on stacked windows, 1968–2012

	CAPM alpha			Four-factor alpha		
State	$t+12$	$t+36$	$t+60$	$t+12$	$t+36$	$t+60$
Low-minus-high (L–H)						
1 (low)	−0.11	0.35	0.25	−0.05	0.28	0.26
	(−0.29)	(0.93)	(0.82)	(−0.18)	(0.86)	(0.89)
2	0.51	0.12	0.29	0.34	−0.01	0.13
	(1.73)	(0.46)	(0.99)	(1.26)	(−0.06)	(0.55)
3	0.34	0.32	0.46	0.04	0.08	0.10
	(1.34)	(1.71)	(2.92)	(0.16)	(0.47)	(0.66)
4	0.40	0.47	0.52	−0.12	−0.22	−0.28
	(1.68)	(1.17)	(1.24)	(−0.67)	(−1.10)	(−1.58)
5 (high)	1.22	1.07	0.94	0.14	0.09	0.06
	(2.26)	(2.93)	(2.65)	(0.45)	(0.39)	(0.25)
Beta neutral (BN)						
1 (low)	−0.12	0.14	0.09	−0.07	0.12	0.13
	(−0.60)	(0.68)	(0.47)	(−0.43)	(0.67)	(0.75)
2	0.15	−0.02	0.11	0.07	−0.07	0.05
	(0.83)	(−0.13)	(0.67)	(0.47)	(−0.61)	(0.33)
3	0.12	0.15	0.25	−0.04	0.01	0.03
	(0.75)	(1.27)	(2.38)	(−0.23)	(0.12)	(0.32)
4	0.24	0.31	0.31	−0.12	−0.10	−0.14
	(1.79)	(1.37)	(1.28)	(−1.08)	(−0.77)	(−1.19)
5 (high)	0.81	0.61	0.52	0.28	0.14	0.08
	(2.63)	(2.71)	(2.49)	(1.39)	(0.83)	(0.54)

Alphas in percent per month. All window-month observations of a state are stacked and estimated in one regression, which equals a weighted regression on calendar months with weights equal to the number of windows covering each month. The estimand is a window-weighted average of monthly abnormal returns with factor loadings common to all windows of the state, not the average of window alphas in Table 3. Newey-West t-statistics (12 lags) in parentheses, computed on the full monthly calendar: months that no window covers enter with a zero score, so that each lag spans calendar months.

Table 16: GFKSW-style t-statistics of the conditional alphas in Table 3, 1968–2012

	CAPM alpha				Four-factor alpha			
	12 months		60 months		12 months		60 months	
State	Ours	GFKSW	Ours	GFKSW	Ours	GFKSW	Ours	GFKSW
Low-minus-high (L–H)								
1 (low)	0.31	0.10	1.54	1.15	0.37	0.31	4.23	1.30
2	2.17	0.90	1.38	1.23	0.59	0.66	1.83	0.72
3	1.11	−0.34	2.62	2.25	1.25	0.16	0.61	−0.67
4	1.60	2.62	2.58	2.69	0.00	0.24	−1.20	−0.58
5 (high)	2.00	5.11	3.60	4.28	0.51	1.86	0.31	1.72
Beta neutral (BN)								
1 (low)	0.11	−0.01	0.98	0.18	0.20	0.19	3.36	0.91
2	1.30	0.44	0.89	0.67	−0.39	0.20	0.99	0.39
3	0.57	−0.51	2.04	2.07	0.76	−0.07	0.13	−0.64
4	1.56	2.56	2.75	2.91	−0.34	0.29	−1.55	−0.53
5 (high)	2.49	5.09	3.14	4.07	1.42	1.85	0.43	1.02

Ours: Newey-West t-statistic with h lags on the sequence of window alphas of one state (formation months of the state in calendar order), our reading of GFKSW's procedure. GFKSW: their published t-statistics (their Table 3).

Table 17: Dependence of the central results on single decades, 1968–2012

	Windows (one year)		State 5 minus state 1		State-5 CAPM alpha	
Decade of formation months left out	State 5	State 1	L–H	BN	L–H	BN
None (all formation months)	108	97	16.7	9.4	0.68	0.52
1960s	108	92	17.5	9.9	0.68	0.52
1970s	91	54	24.0	7.9	0.72	0.55
1980s	60	97	11.6	6.5	0.07	0.14
1990s	80	96	27.4	12.8	1.18	0.78
2000s	93	63	2.4	6.2	0.59	0.50
2010s	108	83	17.4	11.5	0.68	0.52

State assignments are those of the full sample; the formation months of one decade are dropped at a time (the 1960s contain only December 1967 to December 1969). State 5 minus state 1: difference between the average one-year chained excess returns after states 5 and 1 (percentage points per year). CAPM alpha: average window alpha over 12-month windows after state 5 (percent per month).

Table 18: Inference for the central contrast: state 5 minus state 1, one-year returns, 1968–2012

	Low − high			Beta neutral		
Bootstrap	s.e.	t	p	s.e.	t	p
Null model, 12-month blocks	8.5	1.96	0.049	4.6	2.04	0.044
Null model, 24-month blocks (baseline)	8.4	1.99	0.045	4.7	2.02	0.035
Null model, 36-month blocks	8.2	2.03	0.047	4.3	2.20	0.024
Null model, 60-month blocks	7.8	2.14	0.033	4.3	2.18	0.033
Volatility-preserving, 24-month blocks	9.6	1.74	0.076	5.3	1.77	0.071
Formation-month blocks with states, 24 months	9.2	1.82	0.066	4.3	2.20	0.032
Monte Carlo s.e. of the baseline s.e.	0.19			0.10		

Estimates: 16.7 (L–H) and 9.4 (BN) percentage points per year (Table 2). s.e.: standard deviation of the bootstrap draws (percentage points); t = estimate / s.e.; p: share of centred draws at least as large in absolute value as the estimate. Null model: monthly strategy and factor returns resampled jointly in circular blocks, state calendar fixed (Tables 2 and 3 use the 24-month indices). Volatility-preserving: returns standardised by their local volatility (centred 25-month window), resampled with the same indices and rescaled with the local volatility of the original calendar position. Formation-month blocks: blocks of consecutive formation months resampled together with their states and window returns. 999 draws each. Monte Carlo s.e.: delta-method standard error of the bootstrap standard deviation.

Table 19: Simulated size of the tests for state averages and for the central contrast, one-year windows

	State-5 mean: rejections (%)			State 5 − state 1: rejections (%)			Critical value
Test	iid	GARCH	Obs. vol.	iid	GARCH	Obs. vol.	Obs. vol.
Low-minus-high (L–H)							
GFKSW-style Newey-West	14.2	14.6	14.5	–	–	–	–
Null model, 24-month blocks	6.0	5.9	17.2	4.2	4.9	9.3	2.29
Null model, 60-month blocks	6.0	7.4	18.3	4.8	5.3	9.6	2.23
Volatility-preserving, 24-month blocks	5.9	5.5	6.9	4.3	5.2	7.0	2.10
Formation-month blocks, 24 months	12.7	11.0	12.3	10.8	10.8	9.8	2.36
Beta neutral (BN)							
GFKSW-style Newey-West	14.3	12.6	14.9	–	–	–	–
Null model, 24-month blocks	6.0	5.5	10.5	4.2	3.6	5.6	2.03
Null model, 60-month blocks	6.1	6.3	10.7	4.7	4.3	6.6	2.06
Volatility-preserving, 24-month blocks	6.0	5.6	6.0	4.2	4.9	3.8	1.91
Formation-month blocks, 24 months	13.1	10.9	12.2	10.8	10.5	9.2	2.41

Rejection rates of two-sided 5% tests of a true null (the states carry no information) in 1000 simulated samples of 540 months with the actual state calendar of 1968–2012, 299 bootstrap draws per sample. iid: normal returns with the strategy's mean and volatility; GARCH: AR(1)-GARCH(1,1) with Student-t errors, fitted to the strategy by maximum likelihood; Obs. vol.: the fitted AR(1) process with Student-t errors scaled by the fitted conditional volatility of the actual data, so that volatile periods stay where they occurred. Rejections are counted around the simulated mean of each estimator. Critical value: 95% quantile of $|t|$ for the contrast under Obs. vol. (size-adjusted 5% critical value). Monte Carlo standard errors of the rejection rates: at most 1.2 percentage points.

Table 20: Future cumulative returns on the low-minus-high portfolio by B/P spread state, 1930–2012 (replication of GFKSW Table 5)

		This replication						GFKSW	
State	B/P spread	$t+1$	$t+12$	$t+24$	$t+36$	$t+48$	$t+60$	$t+12$	$t+60$
1 (low)	−4.61 to −0.55	−1.20	−12.40	−12.89	−11.50	−10.47	−9.77	[−10.68]	[−8.89]
2	−0.55 to −0.07	0.10	−0.17	0.40	1.01	0.12	−0.67	[−4.57]	[−6.34]
3	−0.07 to 0.10	0.22	2.78	0.51	−0.87	−0.90	−0.48	[0.22]	[0.29]
4	0.10 to 0.26	−0.19	−1.99	−4.31	−3.43	−2.09	−1.19	[0.51]	[−0.79]
5 (high)	0.26 to 0.76	0.52	5.84	3.79	−0.17	−1.21	−1.49	[7.44]	[1.28]

Returns in percent; horizons beyond one month are annualised. States are formed over the formation months December 1929 to November 2012 as in Table 9. GFKSW values in brackets.

B. Additional results

Table 21: Beta estimation frequency and window, 1968–2024 (exploratory)

	Monthly returns			Daily returns			
	M36	M60	M120	D3	D12	D36	DD12
Panel A: beta forecasts							
Q1 beta, ex ante	0.41	0.47	0.53	0.36	0.47	0.52	0.47
Q1 beta, realised	0.63	0.62	0.62	0.63	0.56	0.56	0.56
Q5 beta, ex ante	1.82	1.76	1.68	1.74	1.62	1.57	1.66
Q5 beta, realised	1.42	1.44	1.39	1.38	1.41	1.38	1.46
RMSE, own sort, Q1	0.34	0.26	0.24	0.36	0.24	0.22	0.22
RMSE, own sort, Q5	0.49	0.43	0.38	0.50	0.36	0.33	0.35
RMSE, common assets	0.28	–	0.23	0.26	0.21	0.21	0.20
RMSE of M60, same assets	0.25	–	0.25	0.24	0.24	0.24	0.24
DM t vs. M60	3.63	–	−2.04	1.46	−2.38	−2.92	−3.55
Holm p	0.002	–	0.083	0.14	0.053	0.014	0.002
Panel B: strategies (BN with ex-ante short ratio)							
L–H excess return (%)	−0.5	−0.1	−1.2	−0.5	−1.1	−0.2	−0.9
t-statistic	−0.20	−0.02	−0.47	−0.20	−0.38	−0.07	−0.32
BN excess return (%)	4.4	4.4	3.8	3.6	2.9	3.2	3.5
t-statistic	3.12	3.29	2.86	2.62	2.28	2.29	3.00
BN Sharpe ratio	0.44	0.45	0.38	0.36	0.30	0.32	0.37
ΔSR vs. M60	−0.01	–	−0.08	−0.09	−0.16	−0.14	−0.08
Holm p	1.00	–	0.89	1.00	0.73	0.73	1.00
BN four-factor α (%)	1.0	1.1	0.8	0.0	0.3	0.6	0.8
t-statistic	0.86	0.95	0.69	−0.02	0.24	0.39	0.70
Q1 turnover (% p.a.)	246	179	115	770	291	157	300
BN Sharpe ratio, net	0.40	0.42	0.35	0.26	0.25	0.29	0.32
Valuation effect (pp)	13.5	17.1	12.3	7.5	8.0	11.7	14.4

M36/M60/M120: monthly excess returns over 36, 60 or 120 months (minimum 24, 24, 60); D3/D12/D36: daily excess returns over 3, 12 or 36 months (minimum 50, 190, 560 days); DD12: D12 with one Dimson lag. Same large-cap universe for all. Realised beta: slope of the quintile's monthly excess return on the market. Forecast target: realised beta of a quintile portfolio over the next 12 months, from monthly returns. Own sort: the estimator's own quintiles. Common assets: the quintiles of both sorts in a pair, so both estimators forecast the same portfolios; DM: Diebold-Mariano t-statistic of the squared-error difference (negative = better than M60; Newey-West, 12 lags). Newey-West t-statistics for returns. ΔSR: Ledoit-Wolf test of the BN Sharpe ratio against M60. Net: 10 bp per dollar traded and 25 bp lending fee. Valuation effect: 12-month L–H return after the highest minus after the lowest B/P spread state, with each estimator's own quintiles and states and windows over 1968–2024 (hence not equal to the 1968–2012 contrast of Table 2 for M60). Holm adjustments over the six comparisons with M60 (text: also over all ten forecast comparisons).

Table 22: Diversification and liquidity of the long-only portfolios, 1968–2024

Portfolio	Positions	Largest (%)	Top 10 (%)	Effective N	Size pct.	Volume pct.	Share turn. (%)
Low-beta quintile (Q1)	110	12.3	48	29	0.82	0.74	6.6
Buffer rule	134	10.4	45	34	0.83	0.75	6.7
Quarterly rebalancing	110	12.3	48	29	0.82	0.74	6.6
Capped at 5%	110	5.0	40	44	0.79	0.70	6.9
Daily beta (12 months)	110	11.7	41	38	0.71	0.61	6.7
Minimum variance, monthly	47	3.0	30	38	0.60	0.46	6.4
Minimum variance, daily	82	3.0	28	54	0.47	0.37	8.4
Equal weight ($1/N$)	553	0.2	2	553	0.51	0.50	12.1
Cap weight (benchmark)	553	4.5	22	119	0.79	0.76	9.4

Averages over rebalancing dates of the target weights. Positions: weights of at least 1 bp; effective N: $1/\sum w_i^2$; size and volume percentiles: weight-averaged ranks by market capitalisation and dollar volume within the universe in the formation month; share turnover: monthly volume over shares outstanding (CRSP volume covers at least 99.0% of each portfolio's weight). Similar volume ranks describe the stocks held, not a portfolio's trading cost or capacity, which depend on trade size and market impact.

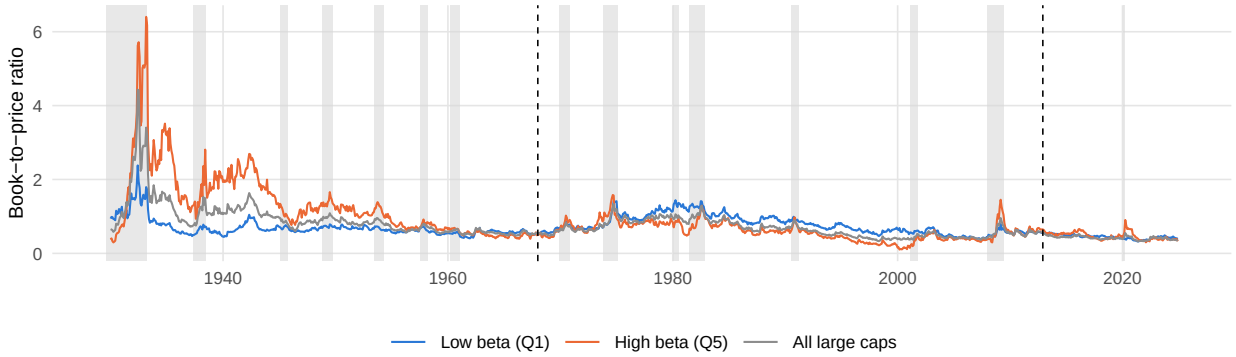


Figure 5: Average book-to-price ratios of the lowest- and highest-beta quintiles and of all large caps, measured at the end of each formation month from December 1929 to November 2024 (their spread is shown in Figure 2). Stock-level B/P ratios are trimmed cross-sectionally at 0.5% in each tail. Shaded areas are NBER recessions; dashed lines mark the first and last formation month of the GFKSW main sample (December 1967, November 2012). Book equity before Compustat coverage is from Davis, Fama and French (2000).

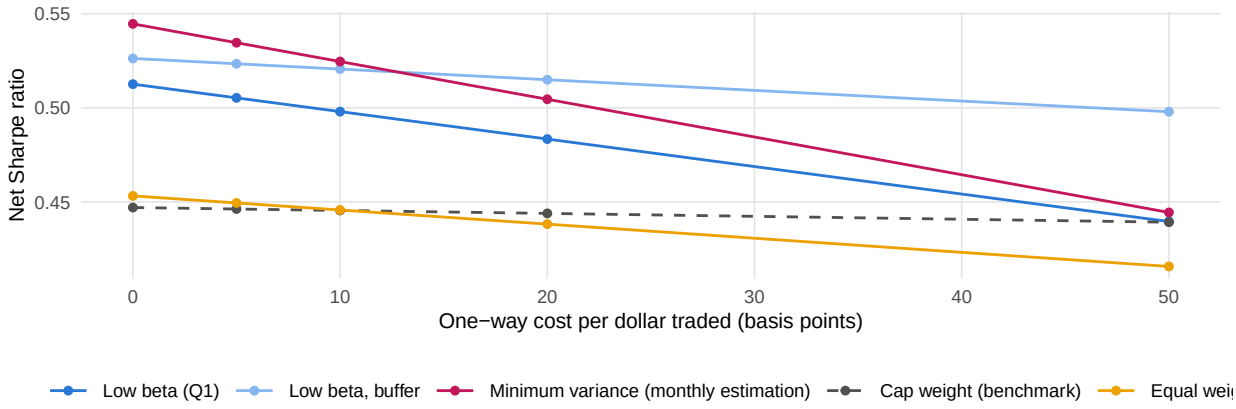


Figure 6: Net Sharpe ratio as a function of the one-way trading cost per dollar traded, 1968–2024, monthly rebalancing (buffer rule as described in Section 7). Costs are charged on each portfolio's turnover, including the cap-weighted benchmark's. All cost levels are scenarios, not estimates of actual costs; break-even costs above 50 basis points quoted in the text extrapolate beyond the range shown.

Table 23: Long-short low-beta strategies before and after trading and lending costs

	Gross						Net of costs			
Portfolio	Excess	SR	Beta	CAPM α	4F α	TO	Excess	SR	CAPM α	4F α
1968–2024										
Low – high	−0.1	0.00	−0.82	5.6	1.9	399	−0.7	−0.04	5.0	1.2
	(−0.02)			(2.58)	(1.01)				(2.28)	(0.66)
Beta neutral (fixed 25%)	4.7	0.47	0.26	2.9	0.8	234	4.4	0.44	2.6	0.5
	(3.50)			(2.21)	(0.74)				(1.98)	(0.47)
Beta neutral (ex ante)	4.4	0.45	0.22	2.8	1.1	251	4.0	0.42	2.5	0.8
	(3.29)			(2.17)	(0.95)				(1.92)	(0.67)
Beta neutral (shrunk beta)	3.1	0.30	−0.06	3.5	1.3	293	2.7	0.26	3.1	0.9
	(2.16)			(2.37)	(0.98)				(2.10)	(0.66)
2013–2024										
Low – high	−7.3	−0.38	−0.80	3.4	−1.4	344	−7.9	−0.41	2.8	−2.0
	(−1.46)			(0.87)	(−0.34)				(0.72)	(−0.49)
Beta neutral (fixed 25%)	5.3	0.55	0.24	2.1	−0.5	181	5.1	0.53	1.9	−0.8
	(2.33)			(0.89)	(−0.23)				(0.80)	(−0.33)
Beta neutral (ex ante)	5.1	0.52	0.22	2.1	−0.6	192	4.8	0.49	1.9	−0.9
	(2.17)			(0.88)	(−0.25)				(0.77)	(−0.35)
Beta neutral (shrunk beta)	1.7	0.16	−0.06	2.5	−0.8	235	1.3	0.12	2.2	−1.1
	(0.61)			(0.91)	(−0.28)				(0.79)	(−0.40)
2016–2024										
Low – high	−11.0	−0.52	−0.84	−0.1	−4.0	353	−11.6	−0.55	−0.7	−4.6
	(−1.81)			(−0.03)	(−0.83)				(−0.16)	(−0.96)
Beta neutral (fixed 25%)	3.2	0.32	0.21	0.4	−1.7	183	3.0	0.29	0.2	−1.9
	(1.23)			(0.16)	(−0.62)				(0.07)	(−0.70)
Beta neutral (ex ante)	2.8	0.27	0.19	0.3	−1.9	196	2.5	0.25	0.1	−2.1
	(1.04)			(0.11)	(−0.66)				(0.02)	(−0.75)
Beta neutral (shrunk beta)	−1.1	−0.09	−0.10	0.2	−2.5	241	−1.4	−0.12	−0.1	−2.8
	(−0.33)			(0.06)	(−0.74)				(−0.05)	(−0.85)

Annualised percentages; Newey-West t-statistics in parentheses; Beta: realised market beta. Net returns deduct 10 bp per dollar traded on both legs and a lending fee of 25 bp per year on the short notional (scenario values); the ex-ante and shrunk-beta versions also pay for changes in the short ratio. Shrunk beta: short ratio $(0.6\hat{\beta}_{Q1} + 0.4)/(0.6\hat{\beta}_{Q5} + 0.4)$ as in Frazzini and Pedersen (2014). TO: annual turnover of both legs in percent of capital.

C. Reproducibility and use of AI tools

All results are produced by build.R, which reads the university database read-only (`tidy_finance.sqlite`, 49.8 GB, last modified 14 May 2026), builds the stock panel, estimates all betas, forms all portfolios and stores every table and time series in results/res.rds; paper.qmd reads only results/res.rds. The results used here were computed on 07 October 2026 with R version 4.4.1 (2024-06-14) and data.table 1.18.2.1, RSQLite 2.4.6, sandwich 3.1.1 and quadprog 1.5.8, with the stock panel built from the database in the same run (run time 13 minutes). Two components, the monthly market betas of Table 7 and the standard errors of Appendix Table 15, were corrected on 09 October 2026 by update_r3.R from these results and the cached daily factors, without rerunning the bootstraps and simulations (build.R contains the same corrections); the document was rendered with R version 4.4.1 (2024-06-14), knitr 1.51 and kableExtra 1.4.0. Random numbers are used only in the bootstraps and the size simulations (seeds derived from 20261006, L'Ecuyer-CMRG; bootstrap indices are drawn in the main process and every simulated sample has its own seed, so results do not depend on the number of cores). A set of checks in paper.qmd stops the rendering if a qualitative statement in the text is no longer supported after a recomputation; the checks never alter a calculation. README.md describes how to run the code and CHANGELOG.md documents the data checks.

Use of AI tools. We used an AI assistant (Anthropic's Claude) to write, run and debug the R code on the university server, to check the results and to draft and revise the text. The authors are responsible for the content.